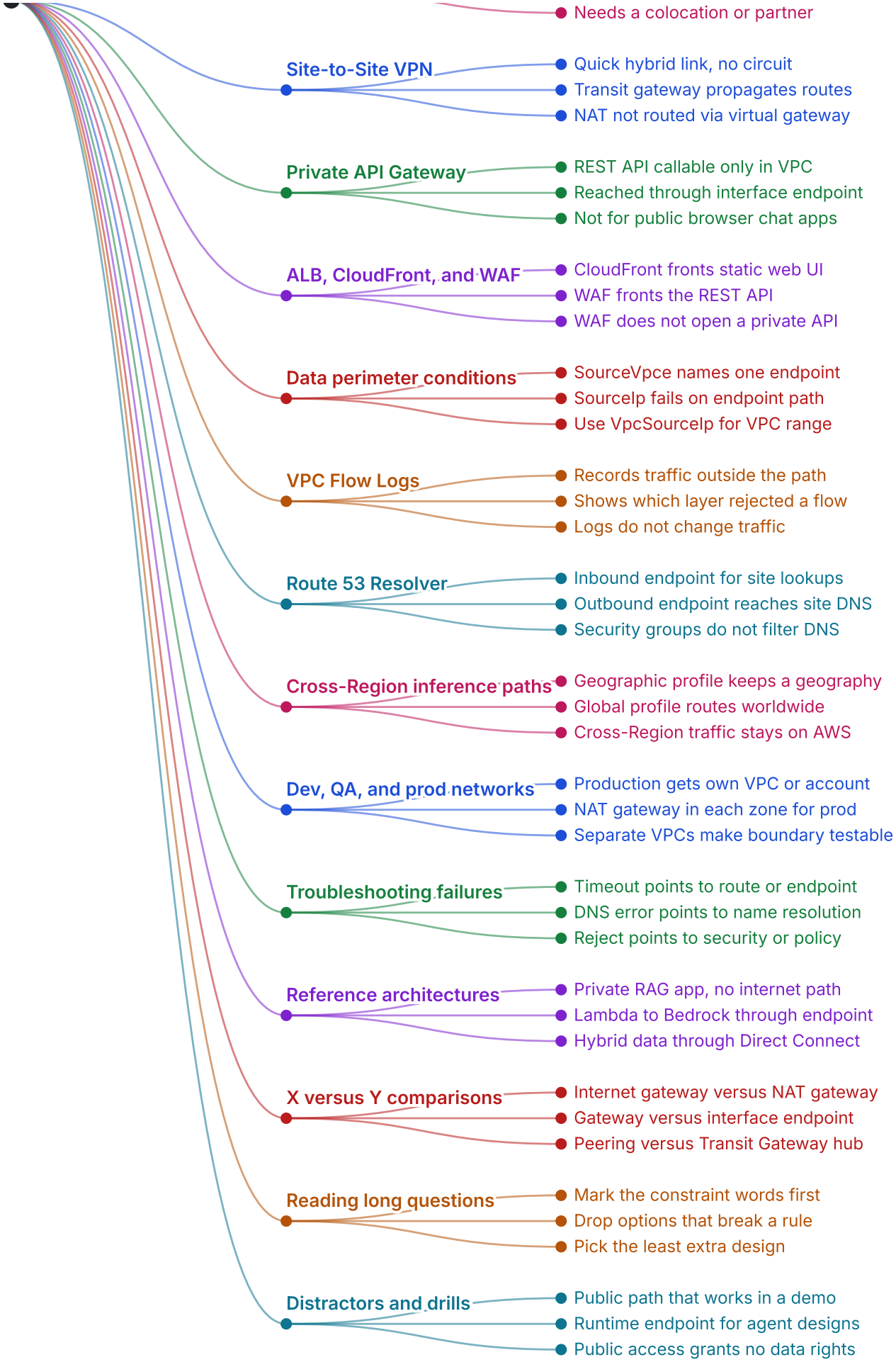
Memory sheet: Networking for generative AI

Cross-domain networking sheet. Read this sheet, then answer the flash questions aloud.


Networking for generative AI
The deciding constraint
Start at the caller, not service
Stop at the first settled branch
Signal words pick the branch
VPC basics for AI workloads
Subnets split the VPC range
Route table sets public or private
Spread tiers across two zones
Internet gateway
Serves public inbound clients
Private callers need no route
Not for private outbound only
NAT gateway
Outbound only for private subnets
One NAT gateway in each zone
Skip NAT when an endpoint exists
Egress-only internet gateway
Outbound IPv6, no new inbound
Not an IPv4 egress fix
Interface endpoints
One endpoint per API surface
Private DNS keeps code names
Agent runtime needs its own name
Gateway endpoints
Route table target, no fee
S3 and DynamoDB only
Not for on-premises callers
Endpoint policies and DNS
Endpoint policy restricts the path
Policy grants no new permissions
Private DNS maps standard names
SageMaker AI in a VPC
Runtime endpoint needed in zone
VPC-only Studio needs endpoints
Default Studio is not private
Lambda in a VPC
VPC attach adds no Bedrock route
Missing route looks like timeout
Attach only for private resources
Private vector store access
AllowFromPublic false blocks public
Network policy is not data access
Bedrock source service on collection
AWS Transit Gateway
Hub for many VPCs and sites
One route table per attachment
Peering suits two VPCs only
VPC peering
One link between two VPCs
Not a hub for many VPCs
No charge to create a link
AWS Direct Connect
Dedicated private link to AWS
Not an encryption feature



Deciding rules

When the stem says	Think of	Not when
Private subnet, no internet, and an AWS API call	Interface endpoint for each API surface, with private DNS	S3 or DynamoDB only, where a gateway endpoint is free
S3 or DynamoDB traffic from one VPC only	Gateway endpoint in the route table	On-premises, peered VPC, or transit gateway callers, which need interface endpoints
A private design calls agents	bedrock-agent-runtime endpoint alongside bedrock-runtime	Direct model calls only, where bedrock-runtime alone fits
On-premises data center or hybrid site	Direct Connect or VPN, Transit Gateway, and Route 53 Resolver	A quick secure link is enough, which points to a Site-to-Site VPN
Many VPCs and an on-premises site share routing	AWS Transit Gateway hub	Two VPCs with one simple link, where peering fits
Residency rule that names a geography	Geographic cross-Region inference profile	Cost matters and no residency rule applies, where a global profile fits
Timeout from a private subnet	Check route table, NAT, and endpoints first	A name error, which is a DNS and resolver check
Browser on the internet must call the backend	Public front door with a private backend behind it	A private REST API, which the internet cannot reach

Etch lines

1. A NAT route still reaches Bedrock over the internet path.
2. Begin with where the caller runs, then take the first settled branch.
3. Gateway endpoints cover S3 and DynamoDB only, never Bedrock.
4. A hang in a private subnet often traces back to a route or endpoint gap.
5. Security groups remember replies, but network ACLs need explicit reply rules.
6. A wider security group cannot fix a subnet ACL that already drops packets.

Flash questions

- Q: A private Lambda function must call a model, and internet access is banned. What is the first fix? A: Add an interface endpoint for bedrock-runtime in the function subnets.
- Q: Which endpoint type adds no extra charge and serves S3 through a route table? A: A gateway endpoint.
- Q: Model calls work from a private subnet, but agent calls fail. What is missing? A: The bedrock-agent-runtime interface endpoint.
- Q: Which gateway lets private subnets start outbound connections but blocks new inbound connections from the internet? A: A NAT gateway.
- Q: Which hub connects many VPCs and an on-premises site? A: AWS Transit Gateway.
- Q: A residency rule keeps inference inside one geography. Which profile type fits? A: A geographic cross-Region inference profile.
- Q: A request is allowed by the security group, but the reply is blocked. Which layer is the likely cause? A: The network ACL, because it is stateless and needs its own reply rule.

- Q: The network path works, but a principal gets a denial on data reads. Which policy do you change?
  A: The data access policy, which decides who may read or write.

Source: frags2 net.html sections, including VPC basics, interface and gateway endpoints, Transit Gateway, Direct Connect, and troubleshooting. AWS docs and the study guide sections above, no practice-test text.